Lecture 8: The Linear Regression Model

notes pp. 55–62

The one-sentence version

The **conditional expectation function** $\mathrm{E}(y \mid x)$ is the best possible predictor of y. OLS fits the best *linear* predictor. The LRM's assumptions make the CEF linear, $\mathrm{E}(y \mid x) = x'\beta$, so OLS estimates β.

1. Prediction: why we care about $\mathrm{E}(y \mid x)$

- With no information, the best guess under squared loss is the mean: $\arg\min_a \mathrm{E}(y-a)^2 = \mathrm{E}(y)$.
- With x, the best guess is the CEF $m(x) = \mathrm{E}(y \mid x)$. Decompose $y = m(x) + u$. By construction $\mathrm{E}(u \mid x) = 0$, so u is uncorrelated with *every* function of x.
- OLS restricts the predictor to be linear, $\hat{y} = x'b$. If the CEF is linear, OLS recovers it. Otherwise OLS gives the best *linear* approximation to it.

2. The OLS mechanics

- Minimize $SSE = \sum(y_i - x_i'b)^2$. The FOCs are $\sum_i x_{ik}e_i = 0$ for all k, or $X'e = 0$. These are the **normal equations**, and solving them gives $b = (X'X)^{-1}X'Y$.
- With a constant: $\sum e_i = 0$, $\bar{y} = \bar{x}'b$ (the line passes through the means), and the residuals are uncorrelated with every regressor.
- Constant only: $b = \bar{y}$. Bivariate: $b_2 = \widehat{\mathrm{Cov}}(x,y)/\widehat{\mathrm{Var}}(x)$ and $b_1 = \bar{y} - b_2\bar{x}$. You get the same answer from the FOCs or from matrix algebra.
- Fit: $R^2 = 1 - SSE/SST$ is the share of the variation in y that the prediction explains.

3. The model: assumptions about the data-generating process

A1 Linearity	$Y = X\beta + \boldsymbol{\varepsilon}$. Linear in the *parameters*, so $\log y$, x^2, $\sin(x)$ are all fine. $\boldsymbol{\varepsilon}$ holds every unobserved determinant of y.
A2 Full rank	No column of X is an exact linear combination of the others, so $(X'X)^{-1}$ exists.
A3 Mean indep.	$\mathrm{E}(\varepsilon_i \mid X) = 0$: knowing X tells you nothing about the average disturbance. This is the *key* assumption.
A4 Spherical errors	$\mathrm{Var}(\boldsymbol{\varepsilon} \mid X) = \sigma^2 I$, i.e. homoskedasticity and no autocorrelation. Covered in L9 and L11.
A5 Normality	$\boldsymbol{\varepsilon} \mid X \sim N(0, \sigma^2 I)$. Needed for exact finite-sample t inference.

A1 + A3 $\Rightarrow \mathrm{E}(y_i \mid x_i) = x_i'\beta$, a linear CEF $\Rightarrow$ OLS estimates β. "Regression" just means conditional expectation.

On the exam (SAQ proofs from this lecture, guide numbers in brackets)

CEF best predictor (mean version, then the $m(x)$ version with $h(x) = m - g$) [P1, P2] · residual facts from the FOCs: $\sum e = 0$, $\bar{y} = \bar{x}'b$, $\widehat{\mathrm{Cov}}(x_k, e) = 0$ [P3] · bivariate b in matrix form, then $b_2 = \widehat{\mathrm{Cov}}/\widehat{\mathrm{Var}}$ [P4].

Lecture 9: Properties of OLS

notes pp. 63–69

The one-sentence version

b is a random variable. Writing $b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$ shows it is **unbiased** (A3), has variance $\sigma^2(X'X)^{-1}$ (A4), is the **best linear unbiased estimator** (Gauss–Markov), and is **consistent and approximately normal** (LLN/CLT).

1. The master equation

$b = (X'X)^{-1}X'(X\beta + \boldsymbol{\varepsilon}) = \beta + \underbrace{(X'X)^{-1}X'}_{A}\boldsymbol{\varepsilon}$. All of the randomness in b comes from $A\boldsymbol{\varepsilon}$, which you can read as the OLS coefficients from regressing $\boldsymbol{\varepsilon}$ on X. Every property below is about $A\boldsymbol{\varepsilon}$.

2. Unbiasedness (needs A1–A3)

Condition on X, which makes A fixed: $\mathrm{E}(b \mid X) = \beta + A\,\mathrm{E}(\boldsymbol{\varepsilon} \mid X) = \beta$. Then the LIE gives $\mathrm{E}(b) = \mathrm{E}_X[\mathrm{E}(b \mid X)] = \beta$. **Intuition:** if X can't predict $\boldsymbol{\varepsilon}$, the noise in b averages out to zero.

3. Variance (adds A4)

- Sandwich: $\mathrm{Var}(b \mid X) = A\mathrm{Var}(\boldsymbol{\varepsilon} \mid X)A' = (X'X)^{-1}X'\,\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X)\,X(X'X)^{-1}$.
- Homoskedasticity ($\mathrm{Var}(\varepsilon_i|X) = \sigma^2$, the diagonal) and no autocorrelation ($\mathrm{Cov}(\varepsilon_i, \varepsilon_j|X) = 0$, the off-diagonal) give $\mathrm{Var}(\boldsymbol{\varepsilon}|X) = \sigma^2 I$, so $\mathrm{Var}(b \mid X) = \sigma^2(X'X)^{-1}$.
- $\mathrm{Var}(b)$ is $K \times K$. The diagonal holds each $\mathrm{Var}(b_k)$. The off-diagonal covariances are nonzero because the coefficients are estimated from the same data.

4. Efficiency: the Gauss–Markov theorem

Under A1–A4, OLS is the **minimum-variance linear unbiased estimator**. Caveats: it only compares estimators that are *linear* in Y and *unbiased*, so a biased estimator can have lower variance. Under heteroskedasticity OLS is still unbiased but no longer guaranteed best.

5. Large samples: OLS is a function of a sample average

- $X'\boldsymbol{\varepsilon} = \sum x_i\varepsilon_i = n\bar{w}$ with $w_i = x_i\varepsilon_i$, so $b = \beta + \left(\frac{X'X}{n}\right)^{-1}\bar{w}$.
- **Consistency (LLN):** $\mathrm{E}(w_i) = 0$ by A3 and the LIE, so $\operatorname{plim}\bar{w} = 0$. Also $\operatorname{plim}\frac{X'X}{n} = \mathrm{E}(x_i x_i')$. Together, $\operatorname{plim} b = \beta$.
- **Normality:** under A5, $b \mid X \sim N(\beta, \sigma^2(X'X)^{-1})$ *exactly*, because a linear function of normals is normal. Without A5, the **CLT** makes $\bar{w}$, and therefore b, approximately normal in large samples. That is why t-tests still work.

Which assumption buys what

A1–A3	unbiased and consistent
+ A4	$\mathrm{Var}(b\|X) = \sigma^2(X'X)^{-1}$, Gauss–Markov (BLUE)
+ A5	exact N and t in finite samples (otherwise approximate, via the CLT)

On the exam (SAQ proofs from this lecture, guide numbers in brackets)

$\mathrm{E}(b|X) = \beta$, then $\mathrm{E}(b) = \beta$ [P5] · $b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$, then the sandwich, then why $\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X) = \mathrm{Var}(\boldsymbol{\varepsilon}|X)$, then $\sigma^2 I$ and $\sigma^2(X'X)^{-1}$ [P6] · $b = \beta + (X'X/n)^{-1}\bar{w}$, plus what the LLN and CLT imply [P7].

Lecture 10: Interpretation of OLS

notes pp. 70–76

The one-sentence version

b_k is the effect of x_k **holding the other regressors fixed**. OLS does this by *partialling out* the other regressors. It cannot hold the unobservables in $\boldsymbol{\varepsilon}$ fixed, so if those are correlated with X, you get **omitted variable bias**.

1. Causal effects vs. OLS coefficients

- In the LRM, $\beta_k = \partial y_i / \partial x_{ik}$ is a causal (marginal) effect: everything else, including ε, is held fixed.
- OLS b_k holds the *other columns of* X fixed, but not ε. Under A1–A3, b_k is still causal on average. If A3 fails, it isn't.

2. Projection and residual-maker matrices

- $P_X = X(X'X)^{-1}X'$: $P_X Y = Xb = \hat{Y}$ (the part of Y predicted by X), and $P_X X = X$.
- $M_X = I - P_X$: $M_X Y = e$ (what's left after removing everything X can predict), and $M_X X = 0$.
- Both are symmetric and idempotent. $P_X e = 0$, which means the residuals are unpredictable by any linear combination of X. Regressing e on X gives all-zero coefficients.
- With a constant $Z = \iota$: P_Z has every entry equal to $1/n$, $P_Z x = \bar{x}$, and $M_Z x = x - \bar{x}$. **Partialling out a constant = demeaning.**

3. Frisch–Waugh–Lovell: how OLS "holds X_2 fixed"

Split $Y = X_1\beta_1 + X_2\beta_2 + \boldsymbol{\varepsilon}$. Multiply by M_2: $Y^* = X_1^*\beta_1 + \boldsymbol{\varepsilon}^*$, where $Y^* = M_2 Y$ and $X_1^* = M_2 X_1$.

1. Remove from Y and X_1 the variation explained by X_2.
2. Regress the leftover Y on the leftover X_1: $b_1 = (X_1^{*\prime} X_1^*)^{-1} X_1^{*\prime} Y^*$. This is the same b_1 as the full regression.

4. Omitted variable bias (OVB)

- Regress Y on X_1 only: $b_1 = \beta_1 + \Gamma\beta_2 + \delta$, with $\Gamma = (X_1'X_1)^{-1}X_1'X_2$ (how X_1 predicts X_2) and $\delta = (X_1'X_1)^{-1}X_1'\boldsymbol{\varepsilon}$.
- **Bias** $= \mathrm{E}(b_1 - \beta_1) = \mathrm{E}(\Gamma)\,\beta_2$. It is zero only if $\beta_2 = 0$ (the omitted variable doesn't matter) or $\mathrm{E}(\Gamma) = 0$ (it isn't predicted by X_1).
- **Signing it** with one omitted Z: $\mathrm{sign}(\mathrm{bias}_k) = \mathrm{sign}(\alpha) \times \mathrm{sign}(\gamma_k)$, where α is Z's effect on y and γ_k is x_k's partial relationship with Z.

5. Why A3 fails, and measurement error

Common causes: **omitted variables**, **simultaneity** (X and Y jointly determined), and **measurement error in** X. Classical measurement error: we observe $x_i = x_i^o + u_i$, so u_i acts as an omitted variable with coefficient $-\beta_2$. Then $\mathrm{E}(b_2) = [1 - \mathrm{E}(\gamma_2)]\beta_2$ with $0 < \mathrm{E}\gamma_2 < 1$. This is **attenuation bias**: the estimate is pulled toward zero.

On the exam (SAQ proofs from this lecture, guide numbers in brackets)

Define P_X, show $P_X H = 0$, say it in words, and show that regressing e on X gives $a = 0$ [P8] $\cdot$ P_Z with a column of ones (dimensions, entries), $P_Z X = \bar{x}$, and $X^* = M_Z X = x_i - \bar{x}$ [P9]. (The FWL and OVB logic also feeds into the L11 multicollinearity question.)

Lecture 11: Inference with OLS

notes pp. 77–86

The one-sentence version

Standardize b_k with its standard error. Because σ^2 must be estimated by $s^2 = e'e/(n-K)$, the standardized estimate is t_{n-K}, which gives CIs, t-tests and p-values. Precision depends on noise σ^2, variation in x_k, and how collinear x_k is with the other regressors.

1. From $N(0,1)$ to t_{n-K}

- $b_k \mid X \sim N(\beta_k, \sigma^2[(X'X)^{-1}]_{kk}) \;\Rightarrow\; \dfrac{b_k - \beta_k}{\sqrt{\sigma^2[(X'X)^{-1}]_{kk}}} \sim N(0,1).$
- Replace the unknown σ^2 with s^2: $\dfrac{b_k - \beta_k}{\widehat{SE}(b_k)} \sim t_{n-K}$, where $\widehat{SE}(b_k) = \sqrt{s^2[(X'X)^{-1}]_{kk}}$.
- **CI:** $b_k \pm \kappa\,\widehat{SE}$, with $\kappa =$ `qt(1-(1-c)/2, n-K)`. Higher confidence means a wider interval.
- **Test** $H_0 : \beta_k = q$: $t = (b_k - q)/\widehat{SE}$, $p = 2[1 - F_{n-K}(|t|)] =$ `2*(1-pt(abs(t), n-K))`. Reject if $p < \alpha$.

2. Why s^2 divides by $n-K$

$e = M_X\varepsilon \Rightarrow e'e = \varepsilon' M_X \varepsilon$ (because M is symmetric and idempotent). Then the trace trick: $\mathrm{E}(e'e|X) = \mathrm{tr}(M_X \mathrm{E}(\varepsilon\varepsilon'|X)) = \sigma^2 \mathrm{tr}(M_X) = \sigma^2(n-K)$. So $s^2 = e'e/(n-K)$ is unbiased. The residuals "use up" K degrees of freedom on estimating b.

3. What drives $\mathrm{Var}(b_k)$

- Bivariate: $\mathrm{Var}(b_2|X) = \dfrac{\sigma^2}{\sum(x_i - \bar{x})^2} = \dfrac{\sigma^2}{(n-1)s_x^2}$. It goes down with less noise, more data, or more spread in x.
- Multiple regression (via FWL): $\boxed{\mathrm{Var}(b_k|X) = \dfrac{\sigma^2}{(1-R_k^2)\,SST_{x_k}}}$, where R_k^2 is the R^2 from regressing x_k on all the *other* regressors.
- $1 - R_k^2$ is the share of x_k's variation that is *unique*, and only that part identifies β_k. $\mathrm{VIF}_k = 1/(1-R_k^2)$.
- Precision improves with $\uparrow SST_{x_k}$ (more variation or larger n), with $\downarrow \sigma^2$ (relevant controls that explain y), and with $\downarrow R_k^2$.

4. Multicollinearity and the bias–variance trade-off

- $R_k^2 \approx 1$ means x_k is nearly a linear combination of the others (close to violating A2), so $\mathrm{Var}(b_k)$ blows up. Symptoms: high R^2 but few significant coefficients, unstable estimates, odd signs.
- The "fix" of dropping the correlated variable lowers the variance, but that variable moves into ε, where it is predictable by x_k. A3 then fails and you get OVB. So you trade lower variance for bias. A larger n eases the tension.

On the exam (SAQ proofs from this lecture, guide numbers in brackets)

Construct the standardized estimate, then $N \to t_{n-K}$ [P10] · $e'e = \varepsilon' M \varepsilon$ and $\mathrm{E}(s^2|X) = \sigma^2$ with no shortcuts [P11] · derive bivariate $\mathrm{Var}(b_2) = \sigma^2/SST_x$ plus the three factors [P12] · write $\mathrm{Var}(b_3) = \sigma^2/[(1-R_3^2)SST]$ and explain each piece [P13] · multicollinearity and why dropping regressors causes bias [P14].

Lecture 12: OLS in Practice

notes pp. 87–92

The one-sentence version

Reading regressions correctly: R^2 vs. adjusted R^2, linear probability models, and especially **log specifications**. Know how to turn each coefficient into a unit, percent, or percentage-point statement, and know when units matter.

1. Fit: R^2 vs. adjusted R^2

- $SST = SSR + SSE$ and $R^2 = SSR/SST = 1 - SSE/SST \in [0, 1]$. OLS minimizes SSE, so it *maximizes* R^2, and adding regressors can never lower it. With small n relative to K this leads to **overfitting**.
- adj-$R^2 = 1 - \dfrac{SSE/(n-K)}{SST/(n-1)} = 1 - \dfrac{\widehat{\mathrm{Var}}(\varepsilon)}{\widehat{\mathrm{Var}}(y)}$. It applies a degrees-of-freedom penalty, can fall when you add a regressor, and can be negative. It works as a proxy for *out-of-sample* fit, while R^2 measures in-sample fit.

2. Linear probability model (binary y)

$\mathrm{E}(y|x) = \Pr(y = 1|x) = x'\beta$. Then b_k is the change in the *probability*, so $100\,b_k$ is in **percentage points**. Example: $\widehat{UNION} = 0.962 - 0.047\,ED$ means one more unit of ED gives a 4.7 pp lower probability of union membership.

3. Logs: derive, don't memorize

Recipe: (i) solve for y in levels, (ii) differentiate with the chain rule, (iii) rearrange into $100\,\partial y/y$ (% change) and/or $100\,\partial x/x$ (1% change).

Model	$\partial y/\partial x$	Read β as	Units matter?	Airline example
level–level	β	+1 unit $x \to +\beta$ units y	yes	
log–level	βy	+1 unit $x \to \approx 100\beta\%$ in y	yes	0.043: +100 mi → 4.3%, +1 mi → 0.043%
level–log	β/x	+1% in $x \to \beta/100$ units of y	no	66: +1% dist → +\$0.66
log–log	$\beta y/x$	+1% in $x \to \beta\%$ in y (elasticity)	no	0.40: +10% dist → +4% fare

Units: rescaling x inside a log only shifts the intercept, since $\log(x/100) = \log x - \log 100$. That's why the level–log and log–log slopes don't depend on units. In log–level, the slope is per unit of x, so you must convert (1 mile = 0.01 hundred-miles).

4. Non-constant marginal effects, and dummies

- Quadratic: $y = \beta_1 + \beta_2 x + \beta_3 x^2$ gives $\partial y/\partial x = \beta_2 + 2\beta_3 x$, which depends on where you evaluate it.
- Interaction: $y = \ldots + \beta_2 x_k + \beta_3 x_j + \beta_4 x_k x_j$ gives $\partial y/\partial x_k = \beta_2 + \beta_4 x_j$.
- Binary x_k: use a discrete change, not a derivative. $\beta_k = \mathrm{E}(y|\cdot, x_k{=}1) - \mathrm{E}(y|\cdot, x_k{=}0)$ under A3. Example: banks in the Atlanta Fed district vs. the St. Louis Fed district (Richardson–Troost).

On the exam (SAQ proofs from this lecture, guide numbers in brackets)

Three airline-fare questions [P15]: log–level (exponentiate, marginal effect $= 0.043\,fare$, per-mile %), level–log (marginal effect $66/dist$, \$0.66 per 1%, units only change the intercept), log–log ($e^{4.25} dist^{0.40}$, marginal effect $0.40\,fare/dist$, 10% → 4%).

Lecture 13: Extensions of the Linear Regression Model

notes pp. 93–102

Disclaimer: may not be on the Unit 2 exam

As of Oct 18, no Canvas announcement says Lecture 13 is on the Unit 2 exam. The Oct 6 announcement points to the Lecture 8–12 SAQs, and no Lecture 13 SAQs have been posted. Lecture 13 was added to the notes on Oct 14 and is included here just in case. Prioritize pages 1–5, and if you hear otherwise in class, trust that.

The one-sentence version

The LRM gets two standard criticisms: the **standard errors are wrong** (A4 fails) and the **estimates are biased** (A3 fails). Fix the first with **robust or clustered SEs**, which keep the same b and change only the SEs. Fix the second with **fixed effects**, which use only within-unit variation.

1. Generalized LRM: drop A4

$Y = X\beta + \boldsymbol{\varepsilon}$, $\mathrm{E}(\boldsymbol{\varepsilon}|X) = 0$, $\mathrm{Var}(\boldsymbol{\varepsilon}|X) = \Sigma$ (any covariance matrix). Consequences: OLS is **still unbiased** (that only needs A1–A3). It is **no longer guaranteed BLUE**. R^2 is still valid. But $\sigma^2(X'X)^{-1}$ is **wrong**, so CIs and tests are wrong. Go back to the sandwich: $\mathrm{Var}(b|X) = (X'X)^{-1}X'\Sigma X(X'X)^{-1}$.

2. Heteroskedasticity-robust (Huber–White) SEs

- Allow $\mathrm{Var}(\varepsilon_i|X) = \sigma_i^2$, keeping no autocorrelation, so $\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$.
- You don't need n separate σ_i^2's. You only need the *middle* term $X'\Sigma X$, which you estimate with squared residuals: $\widehat{\mathrm{Var}}(b|X) = \frac{n}{n-K}(X'X)^{-1}X'\hat{\Sigma}X(X'X)^{-1}$ with $\hat{\Sigma} = \mathrm{diag}(e_1^2, \ldots, e_n^2)$.
- In practice, people often skip testing and just report robust SEs.

3. Breusch–Pagan test for heteroskedasticity

$H_0 : \mathrm{Var}(\varepsilon_i|X) = \sigma^2$. OLS forces $X'e = 0$, so X can't predict e, but it *can* predict e^2. Run the auxiliary regression $e_i^2 = \delta_1 + \delta_2 x_{i2} + \cdots + \delta_K x_{iK} + v_i$. Under H_0, $\delta_2 = \cdots = \delta_K = 0$, so the auxiliary R^2 should be small. Test statistic: $LM = nR_{e^2}^2 \sim \chi^2_{K-1}$. A large LM (small p) means reject homoskedasticity.

4. Clustered SEs (panel data, so drop no-autocorrelation)

- Repeated observations of the same unit share unobservables, so the errors are correlated *within* a unit. Some off-diagonal entries of Σ are then $\neq 0$.
- Allow arbitrary correlation within groups and assume independence across groups. Estimate the middle term as $\sum_{g=1}^{G} X_g'e_g e_g'X_g$, which is block-diagonal with e_ie_j inside each block. Then $\widehat{\mathrm{Var}}(b|X) = \frac{G}{G-1}\frac{n-1}{n-K}(X'X)^{-1}\Big(\sum_g X_g'e_g e_g'X_g\Big)(X'X)^{-1}$.
- **Why not just use $\hat{\Sigma} = ee'$?** Because $X'e = 0$, so $(X'X)^{-1}X'ee'X(X'X)^{-1} = 0$. You *must* restrict the off-diagonal entries.
- "Correct estimates, wrong standard errors": clustering changes the SEs, not b.

5. Fixed effects (FE): attacking omitted variable bias

- There are two ways to hold confounders fixed: (1) add them as controls; (2) use only **within-unit variation**, which holds fixed *all* time-invariant differences between units, observed *or unobserved*.
- Model: $y_{it} = x_{it}'\beta + \alpha_i + \varepsilon_{it}$. Here α_i is a "super control" for everything constant about unit i. Example: $crmrte_{it} = \beta_1 + \beta_2 unem_{it} + \alpha_i + \varepsilon_{it}$.
- Matrix form: $Y = X\beta + D\alpha + \boldsymbol{\varepsilon}$, where D holds the group dummies. By FWL, partialling out D means subtracting **group means** (L10, where projecting on ones gave the mean): $y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)'\beta + (\varepsilon_{it} - \bar{\varepsilon}_i)$. This removes α_i.
- Variation used: FE uses *within*. The between estimator uses *across*-unit averages. Pooled OLS uses *total* (both).
- **Corollaries:** (1) units observed only once contribute nothing, since their demeaned values are 0. (2) Variables that don't vary within a unit can't be included, because they are perfectly collinear with D and violate A2.
- Estimate either with dummies plus OLS (many parameters) or by demeaning (efficient). In R, use `fixest::feols(y ~ x | id)`.

On the exam (SAQ proofs from this lecture, guide numbers in brackets)

No Lecture 13 SAQs exist, so this is a guess. The most proof-like pieces build directly on L9–L11 proofs: (i) the sandwich $\mathrm{Var}(b|X) = (X'X)^{-1}X'\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X)X(X'X)^{-1}$ [P6(b)] with Σ in place of $\sigma^2 I$; (ii) why ee' fails ($X'e = 0$, see [P3], [P8]); (iii) FE as FWL with group dummies, i.e. demeaning [P9].